

ON THE ADMISSIBILITY OF ESTIMATORS OF THE COMMON MEAN OF TWO NORMAL POPULATIONS UNDER SYMMETRIC AND ASYMMETRIC LOSS FUNCTIONS

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Summary: Given a random sample $X_1, \dots, X_m$ from $N(\theta, \sigma_1^2)$ and a random sample $Y_1, \dots, Y_n$ independently from $N(\theta, \sigma_2^2)$, we prove the admissibility of the Best Linear Risk Unbiased Equivariant (BLRUE) estimator of the common mean of the two populations under the squared error and LINEX loss functions. Also the admissibility and inadmissibility of the linear estimators of the form $c\bar{X} + d\bar{Y} + k$ are studied.

1. Introduction

In this paper we consider the problem of estimating the common mean of two normal distributions. The problem of estimating the common mean of two or more independent normal populations has received a great deal of attention for its wide applications in design of experiments, clinical trials and many other real life cases. For Some references in this regard see Graybill and Deal (1959), Sinha and Mouqadem (1982) and Kubokawa (1989). See also Lehmann and Casella (1998) pp. 95–96 for other references and comments.

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In section 2, a class of risk unbiased estimators which combines the means of both samples, i.e. $\bar{X}$ and $\bar{Y}$, is developed and in sections 3 and 4 the region of the inadmissibility and admissibility of the estimators of the form $c\bar{X} + d\bar{Y} + k$ are studied under the squared error loss function

$$L_1(\theta, \hat{\theta}) = (\hat{\theta} - \theta)^2 \quad (1.1)$$

which is a symmetric loss function. In practice, the real loss function is often not symmetric and overestimation of a given amount may lead to more or less serious consequences than underestimation of the same amount. This paper also considers estimation of θ using the LINEX loss function, which is defined as

$$L_2(\theta, \hat{\theta}) = b\{e^{a(\hat{\theta}-\theta)} - a(\hat{\theta} - \theta) - 1\} \quad (1.2)$$

where $b > 0$ is the scale parameter and $a \neq 0$ is the shape parameter.

The LINEX loss function was introduced by Varian (1975) and extensively discussed by Zellner (1986). It is useful when a given positive overestimation error is regarded as more serious than a negative underestimation error of the same magnitude ($a > 0$) or vice versa ($a < 0$). For small values of a , the loss function is close to the squared error loss. See Zellner (1986) for a full discussion of the properties of LINEX loss function, where the importance of employing an appropriate loss function is emphasized.

For application of LINEX loss, see Zellner (1986), Rojo (1987), Sadooghi-Alvandi and Nematollahi (1989), Parsian and Sanjari Farsipour (1993) and Sanjari (1999).

We also derive the class of risk unbiased estimators of the common mean of two normal populations and study the region of the admissibility and

inadmissibility of the estimators of the form $c\bar{X} + d\bar{Y} + k$ under the loss function (1.2) in sections 5 and 6.

2. Risk unbiased equivariant estimation

Consider two independent univariate normal populations distributed as $N(\theta, \sigma_1^2)$ and $N(\theta, \sigma_2^2)$, with $\theta \in R$ and σ_1^2, σ_2^2 known. Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$, be i.i.d. observations from these two populations. From a decision theoretic approach when symmetries are present in a problem, it is natural to require a corresponding symmetry to hold for the estimators. Many statistical estimation problems possess certain natural symmetries. The location parameter estimation problem is an important example. It is symmetric, or, to use the usual terminology, equivariant with respect to translation of the sample space, that is

$$\delta(x_1+r, \dots, x_m+r, y_1+r, \dots, y_n+r) = \delta(x_1, \dots, x_m, y_1, \dots, y_n) + r \quad \text{for all } r \quad (2.1)$$

an estimator satisfying (2.1) will be called equivariant under translation or location equivariant. In a location equivariant estimation problem, if a location equivariant estimator exists which minimizes the constant risk, it is called the minimum risk equivariant (MRE) estimator. Note that an MRE estimator need not be unbiased for general loss functions. Some light is thrown on the possible failure of MRE estimators to be unbiased by considering the following decision theoretic definition of unbiasedness, which depends on the loss function L . Following Lehmann and Casella (1998), an estimator δ of θ is said to be risk-unbiased if it satisfies

$$E_\theta[L(\theta, \delta)] \leq E_\theta[L(\theta', \delta)] \quad \text{for all } \theta' \neq \theta \quad (2.2)$$

where E_θ is expectation w.r.t. density function $f(\cdot | \theta)$. The concept of risk-unbiasedness and its relationship to equivariance were given in Lehmann (1951). If the loss is as in (1.2), (2.2) becomes

$$e^{-a\theta'} E_\theta(e^{a\delta}) - a(E_\theta(\delta) - \theta') - 1 \geq e^{-a\theta} E_\theta(e^{a\delta}) - a(E_\theta(\delta) - \theta) - 1. \quad (2.3)$$

Suppose that $E_\theta(e^{a\delta}) < \infty$. Then, the left side of (2.3) is minimized by

$$\theta' = \frac{1}{a} \log[E_\theta(e^{a\delta})]$$

and the condition of risk-unbiasedness, therefore, reduces to

$$E_\theta[e^{a\delta}] = e^{a\theta}. \quad (2.4)$$

Now note that

$$E[e^{a\bar{X} - \frac{a^2\sigma_1^2}{2m}}] = e^{a\theta} \quad \text{and} \quad E[e^{a\bar{Y} - \frac{a^2\sigma_2^2}{2n}}] = e^{a\theta}.$$

So $\bar{X} - \frac{a\sigma_1^2}{2m}$ and $\bar{Y} - \frac{a\sigma_2^2}{2n}$ are risk unbiased estimators of θ . Now, consider a combined estimator of the form

$$\hat{\theta}_\alpha = \alpha(\bar{X} - \frac{a\sigma_1^2}{2m}) + (1 - \alpha)(\bar{Y} - \frac{a\sigma_2^2}{2n})$$

where $0 < \alpha < 1$ is a real number. Obviously $\hat{\theta}_\alpha$ is a location equivariant estimator. Also note that

$$E[e^{a\hat{\theta}_\alpha + \alpha(1-\alpha)a^2(\frac{\sigma_1^2}{2m} + \frac{\sigma_2^2}{2n})}] = e^{a\theta}$$

and hence

$$\hat{\theta}_{R,U}(\alpha) = \alpha(\bar{X} - \frac{a\sigma_1^2}{2m}) + (1 - \alpha)(\bar{Y} - \frac{a\sigma_2^2}{2n}) + \alpha(1 - \alpha)(\frac{\sigma_1^2}{2m} + \frac{\sigma_2^2}{2n})a$$

is a risk unbiased estimator of θ on the basis of $\bar{X}$ and $\bar{Y}$ under the LINEX loss function (1.2). The risk function of $\hat{\theta}_{R,U}(\alpha)$ with respect to the loss (1.2) is easily computed as

$$R(\hat{\theta}_{R,U}, \theta) = \frac{b\alpha^2\sigma_1^2a^2}{2m} + \frac{b(1 - \alpha)^2\sigma_2^2a^2}{2n}. \quad (2.5)$$

The risk (2.5) is minimized when $\alpha = \frac{m\sigma_2^2}{m\sigma_2^2 + n\sigma_1^2}$ and hence the Best Linear Risk Unbiased Equivariant Estimator (BLRUE) of θ under the LINEX loss is

$$\frac{m\sigma_2^2}{m\sigma_2^2 + n\sigma_1^2} \bar{X} + \frac{n\sigma_1^2}{m\sigma_2^2 + n\sigma_1^2} \bar{Y} - \frac{a\sigma_1^2\sigma_2^2}{2(m\sigma_2^2 + n\sigma_1^2)}. \quad (2.6)$$

With the same approach, the BLRUE estimator of θ under the squared error loss (1.1) is

$$\frac{m\sigma_2^2}{m\sigma_2^2 + n\sigma_1^2} \bar{X} + \frac{n\sigma_1^2}{m\sigma_2^2 + n\sigma_1^2} \bar{Y}. \quad (2.7)$$

(Recall that with squared error as loss, the condition of risk-unbiasedness reduces to the usual unbiasedness condition $E_\theta(\delta) = \theta$.)

Both of the estimators (2.6) and (2.7) are special cases of the more general class of linear estimators of the form $c\bar{X} + d\bar{Y} + k$. To study the admissibility of the estimators (2.6) and (2.7), we study the admissibility of the class of linear estimators of the form $c\bar{X} + d\bar{Y} + k$.

Remark 2.1: In (2.6) and (2.7), the BLRUE estimators for the common mean θ are seen to depend on σ_1^2 and σ_2^2 . When σ_1^2 and σ_2^2 are completely unknown, σ_1^2 and σ_2^2 can be replaced by $S_1^2 = \sum_{i=1}^m (X_i - \bar{X})^2 / (m-1)$ and $S_2^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2 / (n-1)$, respectively. In this case natural and reasonable estimators of θ are provided by

$$\frac{mS_2^2}{mS_2^2 + nS_1^2} \bar{X} + \frac{nS_1^2}{mS_2^2 + nS_1^2} \bar{Y} - \frac{aS_1^2S_2^2}{2(mS_2^2 + nS_1^2)} \quad (2.8)$$

and

$$\frac{mS_2^2}{mS_2^2 + nS_1^2} \bar{X} + \frac{nS_1^2}{mS_2^2 + nS_1^2} \bar{Y}. \quad (2.9)$$

Obviously the estimators (2.8) and (2.9) are location equivariant (see (2.1)) but their risks are complicated.

3. The inadmissibility results under squared error loss (1.1)

Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independently distributed as $N(\theta, \sigma_1^2)$ and $N(\theta, \sigma_2^2)$ respectively, where $\theta \in R$ and σ_1^2, σ_2^2 are known. We shall now prove an inadmissibility result for linear estimators $c\bar{X} + d\bar{Y} + k$, which is quite general and in particular does not require the assumption of normality. The risk function of the estimator $c\bar{X} + d\bar{Y} + k$ with respect to the squared error loss is easily computed as

$$\begin{aligned} \rho(c, d, k) &= E[c\bar{X} + d\bar{Y} + k - \theta]^2 \\ &= \frac{c^2\sigma_1^2}{m} + \frac{d^2\sigma_2^2}{n} + [(c + d - 1)\theta + k]^2. \end{aligned} \quad (3.1)$$

So, we have the following theorem.

Theorem 3.1: The estimator $c\bar{X} + d\bar{Y} + k$ is inadmissible under squared error loss whenever one of the following conditions holds:

- (i) $c > 1$ or $d > 1$
- (ii) $0 < c < 1, 0 < d < 1, c + d > 1$
- (iii) $c < 0, d \leq 1$
- (iv) $c \leq 1, d < 0$
- (v) $c + d < 0$

Proof:

- (i) If $c > 1$, then it follows from (3.1) that

$$\rho(c, d, k) \geq \frac{c^2\sigma_1^2}{m} > \frac{\sigma_1^2}{m} = \rho(1, 0, 0)$$

so that $c\bar{X} + d\bar{Y} + k$ is dominated by $\bar{X}$ and hence is inadmissible.

Similarly, if $d > 1$, then

$$\rho(c, d, k) > \rho(0, 1, 0)$$

so that $c\bar{X} + d\bar{Y} + k$ is dominated by $\bar{Y}$.

- (ii) If $0 < c < 1$, $0 < d < 1$ and $c + d > 1$, then $d^2 > (1 - c)^2$ and hence

$$\begin{aligned}\rho(c, d, k) &\geq \frac{c^2\sigma_1^2}{m} + \frac{d^2\sigma_2^2}{n} \\ &> \frac{c^2\sigma_1^2}{m} + \frac{(1-c)^2\sigma_2^2}{n}.\end{aligned}$$

But the function $g(c) = \frac{c^2\sigma_1^2}{m} + \frac{(1-c)^2\sigma_2^2}{n}$ is minimized when $c = c_0$ where $c_0 = \frac{m\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2}$, also

$$g(c_0) = \frac{\sigma_1^2\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2} = \rho(c_0, d_0, 0)$$

where $d_0 = \frac{n\sigma_1^2}{m\sigma_2^2 + n\sigma_1^2}$. Hence $\rho(c, d, k) > \rho(c_0, d_0, 0)$. Thus, $c\bar{X} + d\bar{Y} + k$ is dominated by $c_0\bar{X} + d_0\bar{Y}$ and hence is inadmissible if $0 < c < 1$, $0 < d < 1$ and $c + d > 1$.

- (iii) If $c < 0$ and $d \leq 1$, then $(c + d - 1)^2 > (d - 1)^2$ and hence

$$\begin{aligned}\rho(c, d, k) &\geq \frac{d^2\sigma_2^2}{n} + [(c + d - 1)\theta + k]^2 \\ &= \frac{d^2\sigma_2^2}{n} + (c + d - 1)^2\left[\theta + \frac{k}{c + d - 1}\right]^2 \\ &> \frac{d^2\sigma_2^2}{n} + (d - 1)^2\left[\theta + \frac{k}{c + d - 1}\right]^2 \\ &= \frac{d^2\sigma_2^2}{n} + [(d - 1)\theta + \frac{(d - 1)k}{c + d - 1}]^2 \\ &= \rho(0, d, \frac{(d - 1)k}{c + d - 1}).\end{aligned}$$

Thus $c\bar{X} + d\bar{Y} + k$ is dominated by the estimator $d\bar{Y} + \frac{(d-1)k}{c+d-1}$.

- (iv) Similarly, if $c \leq 1$ and $d < 0$, then

$$\rho(c, d, k) > \rho(c, 0, \frac{(c-1)k}{c+d-1}).$$

Thus in this case, $c\bar{X} + d\bar{Y} + k$ is dominated by $c\bar{X} + \frac{(c-1)k}{c+d-1}$.

(v) If $c + d < 0$, then $(c + d - 1)^2 > 1$ and hence

$$\begin{aligned}\rho(c, d, k) &\geq [(c + d - 1)\theta + k]^2 = (c + d - 1)^2 \left[\theta + \frac{k}{c + d - 1}\right]^2 \\ &> \left[\theta + \frac{k}{c + d - 1}\right]^2 \\ &= \rho(0, 0, \frac{-k}{c + d - 1}).\end{aligned}$$

Thus, $c\bar{X} + d\bar{Y} + k$ is dominated by the constant estimator $\delta \equiv \frac{-k}{c + d - 1}$.

4. The admissibility results under squared error loss (1.1)

In this section we prove admissibility of the BLRUE estimator of θ under the loss function (1.1). Also the admissibility of the estimators of the form $c\bar{X} + d\bar{Y} + k$ is studied.

Theorem 4.1: The BLRUE estimator of θ under the squared error loss is admissible.

Proof: The notation $\delta(c, d, k)$ is used for $c\bar{X} + d\bar{Y} + k$.

For finding the Bayes estimator of θ , consider the normal prior with mean μ and variance τ^2 . The posterior distribution is then normal with mean and variance given by

$$\frac{\frac{m\bar{X}}{\sigma_1^2} + \frac{n\bar{Y}}{\sigma_2^2} + \frac{\mu}{\tau^2}}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}} \quad \text{and} \quad \frac{1}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}}$$

respectively. It can be seen that the unique Bayes estimator is

$$\frac{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}} \bar{X} + \frac{\frac{n}{\sigma_2^2}}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}} \bar{Y} + \frac{\frac{\mu}{\tau^2}}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}} \quad (4.1)$$

and that the associated Bayes risk is finite and hence the Bayes estimator is admissible.

If $c_0 = \frac{m\sigma_2^2}{m\sigma_2^2 + n\sigma_1^2}$, $d_0 = \frac{n\sigma_1^2}{m\sigma_2^2 + n\sigma_1^2}$ and $k = 0$, the risk of $\delta(c_0, d_0, 0)$ as is seen from (3.1) is given by $\rho(c_0, d_0, 0) = \frac{\sigma_1^2\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2}$. Note that if $c + d = 1$ and $k = 0$, then we have

$$\rho(c, d, k) = \rho(c, 1 - c, 0) = \frac{c^2\sigma_1^2}{m} + \frac{(1 - c)^2\sigma_2^2}{n}. \quad (4.2)$$

It can be shown that the risk (4.2) is minimized when $c = c_0$, and hence $\delta(c, 1 - c, 0) = c\bar{X} + (1 - c)\bar{Y}$ is inadmissible when $c \neq c_0$. To show that $\delta(c_0, d_0, 0)$ is admissible, the limiting Bayes method due to Blyth (1951) may be used. Suppose that $\delta(c_0, d_0, 0)$ is not admissible. Then, there is an estimator δ^* such that

$$\begin{aligned} R(\theta, \delta^*) &\leq R(\theta, c_0\bar{X} + d_0\bar{Y}) \\ &= \frac{\sigma_1^2\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2} \end{aligned}$$

for all θ , and with strict inequality for at least some θ . Now, $R(\theta, \delta)$ is a continuous function of θ for every δ so that there exists $\epsilon > 0$ and $\theta_0 < \theta_1$ such that

$$R(\theta, \delta^*) < \frac{\sigma_1^2\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2} - \epsilon$$

for all $\theta_0 < \theta < \theta_1$. Let r_τ^* be the average risk of δ^* with respect to the prior distribution $N(0, \tau^2)$, and let r_τ be the Bayes risk of the Bayes estimator (4.1) with respect to $N(0, \tau^2)$. Then it follows that

$$r_\tau = \frac{\sigma_1^2\sigma_2^2\tau^2}{m\sigma_2^2\tau^2 + n\sigma_1^2\tau^2 + \sigma_1^2\sigma_2^2}.$$

Hence

$$\begin{aligned} \frac{\frac{\sigma_1^2\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2} - r_\tau^*}{\frac{\sigma_1^2\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2} - r_\tau} &= \frac{\frac{1}{\sqrt{2\pi}\tau} \int_{-\infty}^{+\infty} \left[\frac{\sigma_1^2\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2} - R(\theta, \delta^*) \right] e^{-\theta^2/2\tau^2} d\theta}{\frac{\sigma_1^2\sigma_2^2}{n\sigma_1^2 + m\sigma_2^2} - \frac{\sigma_1^2\sigma_2^2\tau^2}{m\sigma_2^2\tau^2 + n\sigma_1^2\tau^2 + \sigma_1^2\sigma_2^2}} \end{aligned}$$

$$\geq \frac{(n\sigma_1^2 + m\sigma_2^2)(m\sigma_2^2\tau^2 + n\sigma_1^2\tau^2 + \sigma_1^2\sigma_2^2)\epsilon}{\tau\sqrt{2\pi}\sigma_1^4\sigma_2^4} \int_{\theta_0}^{\theta_1} e^{-\theta^2/2\tau^2} d\theta.$$

The integrand converges monotonically to 1 as $\tau \rightarrow \infty$. By the Lebesgue monotone convergence theorem, the integral therefore converges to $\theta_1 - \theta_0$ and hence the ratio converges to ∞ . Thus, there exists $\tau_0 < \infty$ such that $r_{\tau_0^*} < r_{\tau_0}$, which contradicts the fact that r_{τ_0} is the Bayes risk for $N(0, \tau_0^2)$. It follows that $\delta(c_0, d_0, 0)$ is admissible.

Note that if $c = 0$, $d = 0$, then $\delta(0, 0, k)$ is admissible since it is the only estimator with zero risk at $\theta = k$. Also from (4.1) the coefficients of $\bar{X}$ and $\bar{Y}$ are strictly between 0 and 1, and their sum is strictly less than 1. It is conjectured that $c\bar{X} + d\bar{Y} + k$ is admissible whenever $0 \leq c < 1$, $0 \leq d < 1$, $0 \leq c + d < 1$.

Remark 4.1: It is seen that the estimator (2.7) is the limit of the Bayes estimator relative to the normal prior when $\tau \rightarrow \infty$. Hence the BLRUE estimator of θ under the squared error loss is the generalized Bayes estimator relative to a diffuse prior $\pi(\theta) = 1$ for all $\theta \in R$.

5. The inadmissibility results under LINEX loss (1.2)

Consider the question of inadmissibility of the estimators of the form $c\bar{X} + d\bar{Y} + k$ under the loss (1.2). Since the parameter b does not have any influence on our results without loss of generality we can take $b = 1$. The risk function of the estimator $c\bar{X} + d\bar{Y} + k$ with respect to the loss (1.2) is easily computed as

$$\begin{aligned} \rho(c, d, k) &= E[e^{a(c\bar{X} + d\bar{Y} + k - \theta)} - a(c\bar{X} + d\bar{Y} + k - \theta) - 1] \quad (5.1) \\ &= e^{\frac{a^2}{2}(\frac{c^2\sigma_1^2}{m} + \frac{d^2\sigma_2^2}{n}) + a(c+d-1)\theta + ak} - a(c + d - 1)\theta - ak - 1. \end{aligned}$$

So, we have the following theorem.

Theorem 5.1: The linear estimator $c\bar{X} + d\bar{Y} + k$ is inadmissible under LINEX loss function (1.2) whenever one of the following conditions holds:

- (i) $c > 1$ or $d > 1$
- (ii) $0 < c < 1$, $0 < d < 1$, $c + d > 1$
- (iii) $c < 0$ or $d < 0$

Proof:

(i) If $c > 1$, then

$$\begin{aligned} \rho(c, d, k) &= e^{\frac{a^2}{2}(\frac{c^2\sigma_1^2}{m} + \frac{d^2\sigma_2^2}{n}) + a(c+d-1)\theta + ak} - a(c+d-1)\theta - ak - 1 \\ &\geq \frac{a^2}{2}\left(\frac{c^2\sigma_1^2}{m} + \frac{d^2\sigma_2^2}{n}\right) \quad (\text{since } e^x \geq 1 + x) \\ &\geq \frac{a^2c^2\sigma_1^2}{2m} \\ &> \frac{a^2\sigma_1^2}{2m} = \rho(1, 0, \frac{-a\sigma_1^2}{2m}). \end{aligned}$$

So that $c\bar{X} + d\bar{Y} + k$ is dominated by $\bar{X} - \frac{a\sigma_1^2}{2m}$.

Similarly, if $d > 1$, then

$$\rho(c, d, k) > \rho(0, 1, \frac{-a\sigma_2^2}{2n}).$$

So that $c\bar{X} + d\bar{Y} + k$ is dominated by $\bar{Y} - \frac{a\sigma_2^2}{2n}$.

(ii) If $0 < c < 1$, $0 < d < 1$ and $c + d > 1$, then $d^2 > (1 - c)^2$ and

hence

$$\begin{aligned} \rho(c, d, k) &\geq \frac{a^2c^2\sigma_1^2}{2m} + \frac{a^2d^2\sigma_2^2}{2n} \\ &> \frac{a^2c^2\sigma_1^2}{2m} + \frac{a^2(1-c)^2\sigma_2^2}{2n}. \end{aligned}$$

But the function $g(c) = \frac{a^2c^2\sigma_1^2}{2m} + \frac{a^2(1-c)^2\sigma_2^2}{2n}$ is minimized at $c_0 = \frac{m\sigma_2^2}{m\sigma_2^2 + n\sigma_1^2}$, and $g(c_0) = \frac{a^2\sigma_1^2\sigma_2^2}{2(m\sigma_2^2 + n\sigma_1^2)} = \rho(c_0, d_0, k_0)$, where $d_0 = \frac{n\sigma_1^2}{m\sigma_2^2 + n\sigma_1^2}$ and $k_0 = \frac{-a\sigma_1^2\sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)}$. Hence $\rho(c, d, k) > \rho(c_0, d_0, k_0)$.

Thus, $c\bar{X} + d\bar{Y} + k$ is dominated by $c_0\bar{X} + d_0\bar{Y} + k_0$.

- (iii) If $c < 0$, then it will be shown that $c\bar{X} + d\bar{Y} + k$ is dominated by $d^*\bar{Y} + k^*$ where $d^* = \frac{d}{1-c}$, $k^* = \frac{k}{1-c} + \frac{acd^2\sigma_2^2}{2n(1-c)^3}$. Since $e^{\frac{a^2c^2\sigma_1^2}{2m}} > 1$, we have from (5.1):

$$\rho(c, d, k) - \rho(0, d^*, k^*) > e^{\frac{a^2d^2\sigma_2^2}{2n} + a(c+d-1)\theta + ak} - e^{\frac{a^2d^{*2}\sigma_2^2}{2n} + a(d^*-1)\theta + ak^*} - ac\theta - a(d-d^*)\theta - a(k-k^*).$$

Now, using the inequality

$$e^x - e^y \geq (x-y)e^y \quad \text{for all } x, y,$$

and noting that $d^2 - d^{*2} = \frac{c(c-2)d^2}{(1-c)^2} \geq 0$, it follows that

$$\rho(c, d, k) - \rho(0, d^*, k^*) > [ac\theta + a(d-d^*)\theta + a(k-k^*)]e^{\frac{a^2d^{*2}\sigma_2^2}{2n} + a(d^*-1)\theta + ak^*} - ac\theta - a(d-d^*)\theta - a(k-k^*).$$

But $d - d^* = -cd^*$ and $k - k^* = -ck^* - \frac{acd^{*2}\sigma_2^2}{2n}$, hence

$$\begin{aligned} \rho(c, d, k) - \rho(0, d^*, k^*) &> [ac\theta - acd^*\theta - ack^* - \frac{ca^2d^{*2}\sigma_2^2}{2n}] \\ &\quad \times e^{\frac{a^2d^{*2}\sigma_2^2}{2n} + ad^*\theta + ak^* - a\theta} \\ &\quad - ac\theta + acd^*\theta + ack^* + \frac{ca^2d^{*2}\sigma_2^2}{2n} \\ &= -c[\frac{a^2d^{*2}\sigma_2^2}{2n} + ad^*\theta + ak^* - a\theta] \\ &\quad \times [e^{\frac{a^2d^{*2}\sigma_2^2}{2n} + ad^*\theta + ak^* - a\theta} - 1] \\ &\geq 0. \end{aligned}$$

Since $c < 0$ and $y(e^y - 1) \geq 0$, for all y .

If $d < 0$, then it can be shown that $c\bar{X} + d\bar{Y} + k$ is dominated by $c^*\bar{X} + k^*$ where $c^* = \frac{c}{1-d}$ and $k^* = \frac{k}{1-d} + \frac{adc^2\sigma_1^2}{2m(1-d)^3}$.

6. The admissibility results under LINEX loss (1.2)

In this section we prove admissibility of the estimator (2.6). Also the admissibility of the estimators of the form $c\bar{X} + d\bar{Y} + k$ is studied under LINEX loss.

Theorem 6.1: The BLRUE estimator of θ under the LINEX loss is admissible.

Proof: The notation $\delta(c, d, k)$ is used for $c\bar{X} + d\bar{Y} + k$.

Consider the Bayes estimator when the prior distribution of θ is normal with mean μ and variance τ^2 . Then, using (3.2) in Zellner (1986), it follows that the unique Bayes estimator is

$$\frac{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}} \bar{X} + \frac{\frac{n}{\sigma_2^2}}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}} \bar{Y} - \frac{1}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2} + \frac{1}{\tau^2}} \left(\frac{a}{2} - \frac{\mu}{\tau^2} \right) \quad (6.1)$$

and that the associated Bayes risk is finite and hence the Bayes estimator is admissible.

If $c_0 = \frac{m\sigma_2^2}{m\sigma_2^2 + n\sigma_1^2}$, $d_0 = \frac{n\sigma_1^2}{m\sigma_2^2 + n\sigma_1^2}$ and $k_0 = \frac{-a\sigma_1^2\sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)}$, the risk of $\delta(c_0, d_0, k_0)$ as is seen from (5.1) is given by

$$\rho(c_0, d_0, k_0) = \frac{a^2\sigma_1^2\sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)}.$$

Note that if $c + d = 1$, then we have

$$\rho(c, d, k) = \rho(c, 1 - c, k) = e^{\frac{a^2c^2\sigma_1^2}{2m} + \frac{a^2(1-c)^2\sigma_2^2}{2n} + ak} - ak - 1. \quad (6.2)$$

It can be shown that the risk (6.2) is minimized when $c = c_0$ and $k = k_0$, and hence $\delta(c, 1 - c, k) = c\bar{X} + (1 - c)\bar{Y} + k$ is inadmissible when $c \neq c_0$ and $k \neq k_0$.

To show that $\delta(c_0, d_0, k_0)$ is admissible, we again use Blyth's technique. Suppose that $\delta(c_0, d_0, k_0)$ is not admissible, then there exists an estimator δ^*

such that

$$\begin{aligned} R(\theta, \delta^*) &\leq R(\theta, c_0 \bar{X} + d_0 \bar{Y} + k_0) \\ &= \frac{a^2 \sigma_1^2 \sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)} \end{aligned}$$

for all θ , and with strict inequality for at least some θ . By the continuity of $R(\theta, \delta^*)$, there exists $\epsilon > 0$ and $\theta_0 < \theta_1$ such that

$$R(\theta, \delta^*) < \frac{a^2 \sigma_1^2 \sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)} - \epsilon$$

for all $\theta_0 < \theta < \theta_1$. Let r_τ^* be the average risk of δ^* with respect to the prior distribution $N(0, \tau^2)$, and let r_τ be the Bayes risk of the Bayes estimator (6.1) with respect to $N(0, \tau^2)$. Then, by the relation (5.1), it can be shown that

$$r_\tau = \frac{a^2 \sigma_1^2 \sigma_2^2 \tau^2}{2(m\sigma_2^2 \tau^2 + n\sigma_1^2 \tau^2 + \sigma_1^2 \sigma_2^2)}$$

and hence

$$\begin{aligned} \frac{\frac{a^2 \sigma_1^2 \sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)} - r_\tau^*}{\frac{a^2 \sigma_1^2 \sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)} - r_\tau} &= \frac{\frac{1}{\sqrt{2\pi}\tau} \int_{-\infty}^{+\infty} [\frac{a^2 \sigma_1^2 \sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)} - R(\theta, \delta^*)] e^{-\theta^2/2\tau^2} d\theta}{\frac{a^2 \sigma_1^2 \sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)} - \frac{a^2 \sigma_1^2 \sigma_2^2 \tau^2}{2(m\sigma_2^2 \tau^2 + n\sigma_1^2 \tau^2 + \sigma_1^2 \sigma_2^2)}} \\ &\geq \frac{2(n\sigma_1^2 + m\sigma_2^2)(m\sigma_2^2 \tau^2 + n\sigma_1^2 \tau^2 + \sigma_1^2 \sigma_2^2)\epsilon}{\tau \sqrt{2\pi} a^2 \sigma_1^4 \sigma_2^4} \int_{\theta_0}^{\theta_1} e^{-\theta^2/2\tau^2} d\theta. \end{aligned}$$

Now, as $\tau \rightarrow \infty$, Blyth's ratio converges to ∞ , i.e.

$$\frac{\frac{a^2 \sigma_1^2 \sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)} - r_\tau^*}{\frac{a^2 \sigma_1^2 \sigma_2^2}{2(n\sigma_1^2 + m\sigma_2^2)} - r_\tau} \rightarrow \infty$$

and this completes the proof.

If $c = 0$, $d = 0$, then $\delta(0, 0, k)$ is admissible since it is the only estimator with zero risk at $\theta = k$. Also from (6.1) the coefficients of $\bar{X}$ and $\bar{Y}$ are strictly between 0 and 1, and their sum is strictly less than 1. It is conjectured

that $c\bar{X} + d\bar{Y} + k$ is admissible whenever $0 \leq c < 1$, $0 \leq d < 1$, $0 \leq c + d < 1$.

Remark 6.1: Also notice that the estimator (2.6) is the limit of the Bayes estimator (6.1) as $\tau \rightarrow \infty$. Hence it is the generalized Bayes estimator relative to a diffuse prior $\pi(\theta) = 1$ for all $\theta \in R$.

Remark 6.2: The Bayes estimators given in (4.1) and (6.1) are minimax since their risks are constant.

7. Conclusions

We consider the problem of estimating the common mean of two normal populations under the squared error and LINEX loss functions. A general form of the linear risk unbiased equivariant estimators is derived. Under the squared error and LINEX loss functions, conditions for c , d and k are obtained such that a class of linear estimators of the form $c\bar{X} + d\bar{Y} + k$ is admissible or inadmissible. Further study of the properties of alternative estimators in the case where the variances are unknown, would be useful and is left for future research.

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